Gravitational Pressure waves in PPC law of Gravity and Equation of Gravitational Pressure waves

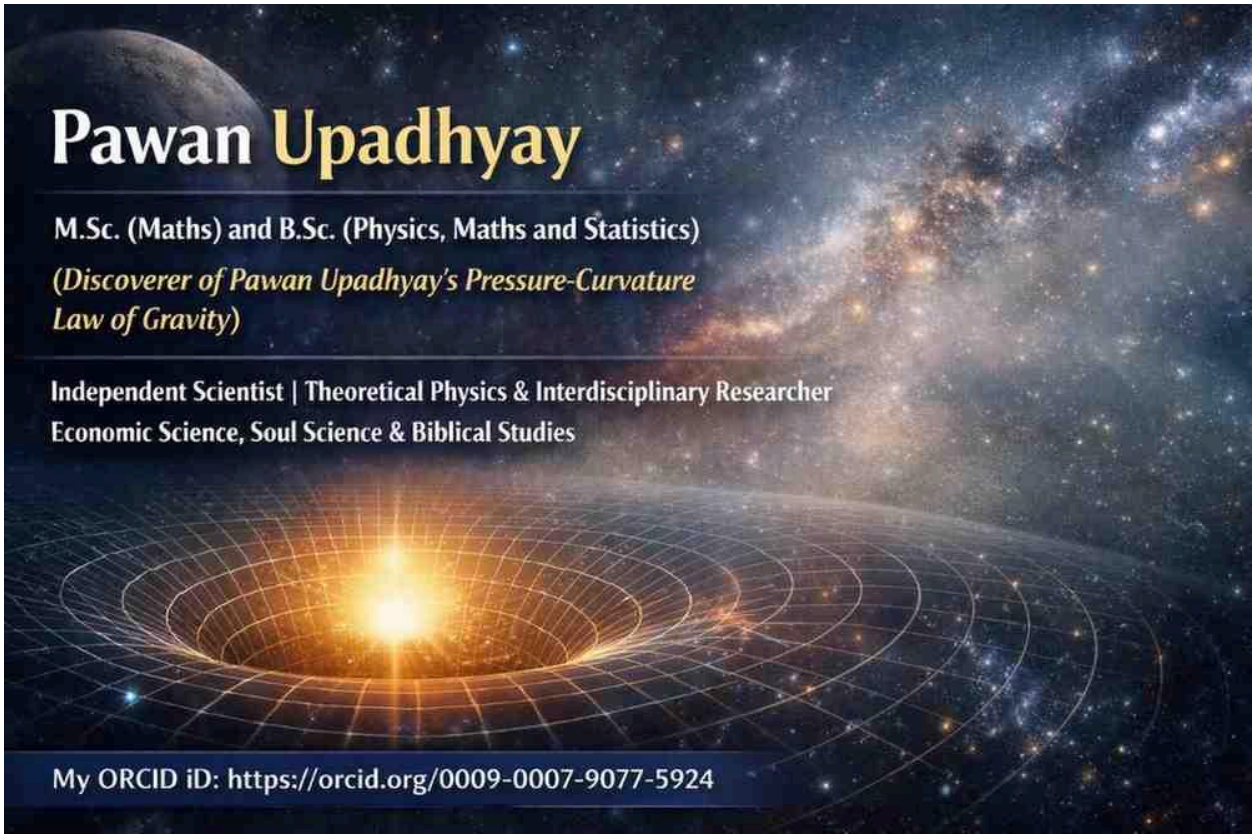
🪐 PPC LAW OF GRAVITY

(Pawan Upadhyay's Pressure–Curvature Law of Gravity)

Pressure Waves in the PPC Law of Gravity

Author: *Pawan Upadhyay (Independent Researcher, 2025)*

📧 *pawanupadhyay28@hotmail.com*


Pawan Upadhyay
M.Sc. (Maths) and B.Sc. (Physics, Maths and Statistics)
(Discoverer of Pawan Upadhyay's Pressure-Curvature Law of Gravity)
Independent Scientist | Theoretical Physics & Interdisciplinary Researcher
Economic Science, Soul Science & Biblical Studies
My ORCID iD: https://orcid.org/0009-0007-9077-5924

1. Introduction

In the framework of the **Pressure–Curvature Law of Gravity (PPC Law)**, gravity is interpreted as a manifestation of **pressure-induced curvature** within spacetime.
Mass not only bends spacetime through its energy density but also applies a **pressure field** upon it.
When this pressure varies dynamically, **Pressure Waves** arise — oscillations that propagate through the universal pressure field at the speed of light.

These waves are the physical expression of what General Relativity describes as gravitational waves.
Thus, *Pressure Waves* form the **dynamic extension** of the PPC Law of Gravity.

My Symbol for Energy density is E_d and for gravitational pressure is P_g.
Where $P_g=\omega E_d$, if $\omega=1$ then $P_g=E_d$.

2. Theoretical Foundation

The PPC Law establishes that:

$$P_g = \rho c^2$$

The **pressure gradient** gives rise to a gravitational force:

$$F = \nabla P_g$$

In Einstein's formulation:

$$G_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}$$

$$T_{\mu\nu} = \left(\rho + \frac{p}{c^2}\right) u_\mu u_\nu + p g_{\mu\nu}$$

In the weak-field limit:

$$\nabla^2 \Phi = 4\pi G(\rho + 3p/c^2)$$

PPC Equations:
Modified GR equations,

$$T_{\mu\nu} = (E_d + P_g)u_\mu u_\nu + P_g g_{\mu\nu}$$

$$G_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}$$

Substituting PPC tensor:

$$G_{\mu\nu} = \frac{8\pi G}{c^4} \left[(E_d + P_g)u_\mu u_\nu + P_g g_{\mu\nu}\right]$$

$$\nabla^2 \Phi = 4\pi G \left(\rho + \frac{3P_g}{c^2}\right)$$

Using:

$$\rho = \frac{E_d}{c^2}$$

We obtain:

$$\nabla^2 \Phi = \frac{4\pi G}{c^2}(E_d + 3P_g)$$

3. Linearized Gravitational Pressure Wave Equation

In weak-field approximation:

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}, \quad |h_{\mu\nu}| \ll 1$$

Define the trace-reversed perturbation:

$$\bar{h}_{\mu\nu} = h_{\mu\nu} - \tfrac{1}{2}\eta_{\mu\nu} h$$

Impose the harmonic (Lorenz) gauge condition:

$$\partial^{\mu} \bar{h}_{\mu\nu} = 0$$

In vacuum, Einstein's equations reduce to:

$$\boxed{\Box \bar{h}_{\mu\nu} = 0}$$

Where the d'Alembert operator is:

$$\Box = \frac{1}{c^2}\frac{\partial^2}{\partial t^2} - \nabla^2$$

Expanded form:

$$\boxed{\frac{\partial^2 \bar{h}_{\mu\nu}}{\partial t^2} = c^2 \nabla^2 \bar{h}_{\mu\nu}}$$

This is the gravitational wave equation.

In PPC interpretation:

Dynamic gravitational pressure
→ Dynamic curvature
→ Pressure–curvature wave propagation

Gravitational Pressure waves and Mechanical Pressure waves are two different waves.

Formation of Pressure Waves :-

3. Formation of Pressure Waves

If the pressure or density field varies with time:

$$p = p(x,t), \quad \rho = \rho(x,t)$$

These are described by the classical wave equation:

$$\frac{\partial^2 \Phi}{\partial t^2} = c^2 \nabla^2 \Phi$$

4. Flow of PPC Law

Mass → Pressure → Force → Curvature → Pressure Waves in Spacetime

Figure 1 — Propagation of Pressure Waves according to the PPC Law
Pawan Upadhyay — Discoverer of the PPC Law of Gravity (2025)

5. Physical Interpretation

Pressure Waves are **vibrations of the spacetime pressure field** created by mass-energy variations.
They carry curvature energy through the universe and manifest as detectable gravitational waves.

Under this interpretation:

- Gravity arises from **pressure-induced curvature**.
- Gravitational waves are **pressure waves** in spacetime.
- The PPC Law unifies mass, pressure, and curvature in a single causal chain.

6. Summary

> “Gravitational waves are pressure waves in spacetime — vibrations of the universal pressure field created by mass.”
> — *Pawan Upadhyay, 2025*

7. References

1. Einstein, A. (1915). *The Field Equations of Gravitation. Sitzungsberichte der Preussischen Akademie der Wissenschaften.*

2. Abbott, B. P. *et al.* (LIGO Scientific Collaboration) (2016). *Observation of Gravitational Waves from a Binary Black Hole Merger. Physical Review Letters*, 116, 061102.

3. Upadhyay, P. (2025). *PPC Law of Gravity — Independent Discovery.* GitHub Repository: https://github.com/pawanupadhyay2025/PPC-law-of-gravity-

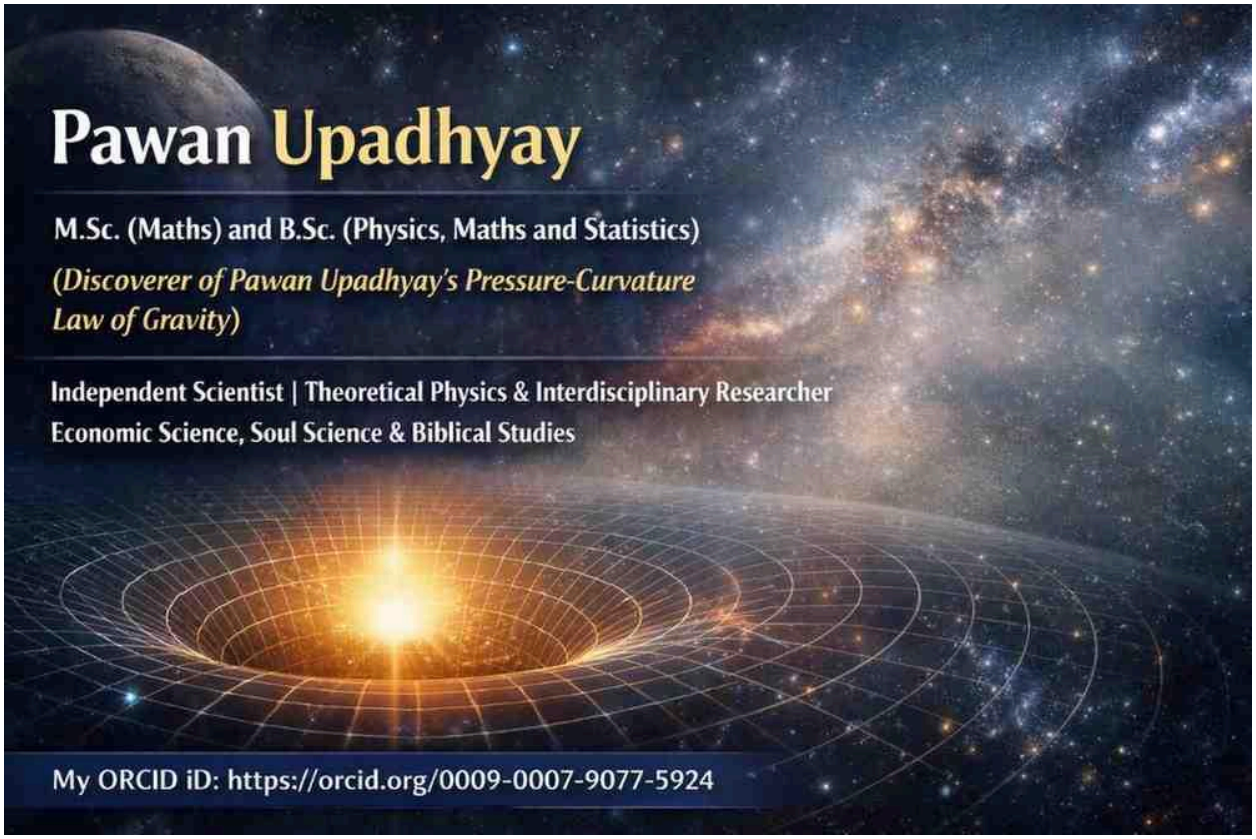
✔️ Paper Summary

This paper presents a new interpretation of gravitational waves as *Pressure Waves* — dynamic oscillations in spacetime's pressure field.
It extends Einstein's General Relativity by identifying pressure as the **physical mechanism** behind curvature formation, completing the causal sequence:
Mass → Pressure → Force → Curvature → Pressure Waves → Motion.

[Expanded Research Paper]

📄 The Classical Wave Equation: Foundations, Derivation, and Physical Interpretations

Author: Pawan Upadhyay
Affiliation: Independent Researcher
Email: pawanupadhyay28@hotmail.com



Abstract

The classical wave equation is a fundamental partial differential equation that governs the propagation of disturbances in physical systems. It appears across multiple domains including acoustics, elasticity, and field theory. This paper presents a rigorous formulation of the classical wave equation, its derivation from physical principles, analytical solutions, and its broader implications in physics. Special emphasis is placed on its interpretation as a universal propagation law independent of specific physical realization.

---

1. Introduction

Wave phenomena are ubiquitous in nature, appearing in systems ranging from vibrating strings to electromagnetic radiation. At the heart of these phenomena lies a unifying mathematical structure known as the **classical wave equation**:

$$\frac{\partial^2 \Phi}{\partial t^2} = c^2 \nabla^2 \Phi$$

This equation describes how a scalar field $\Phi(\mathbf{r}, t)$ evolves in space and time.

The goal of this paper is to:

- Establish the mathematical structure of the wave equation
- Derive it from physical principles
- Analyze its solutions
- Clarify its physical interpretations

The goal of this paper is to:

- Establish the mathematical structure of the wave equation
- Derive it from physical principles
- Analyze its solutions
- Clarify its physical interpretations

---

2. Mathematical Formulation

The classical wave equation in three dimensions is:

2. Mathematical Formulation

The classical wave equation in three dimensions is:

$$\frac{\partial^2 \Phi}{\partial t^2} = c^2 \left(\frac{\partial^2 \Phi}{\partial x^2} + \frac{\partial^2 \Phi}{\partial y^2} + \frac{\partial^2 \Phi}{\partial z^2} \right)$$

or compactly:

$$\frac{\partial^2 \Phi}{\partial t^2} = c^2 \nabla^2 \Phi$$

Key Properties

- Linear
- Second-order PDE
- Hyperbolic type
- Admits wave-like solutions

Key Properties

- Linear
- Second-order PDE
- Hyperbolic type
- Admits wave-like solutions

---

3. Physical Derivation

3.1 From a Vibrating String

3. Physical Derivation

3.1 From a Vibrating String

Consider a string under tension T with linear density μ. Let $u(x,t)$ represent transverse displacement.

Using Newton's second law:

$$\mu \frac{\partial^2 u}{\partial t^2} = T \frac{\partial^2 u}{\partial x^2}$$

Rewriting:

$$\frac{\partial^2 u}{\partial t^2} = \frac{T}{\mu} \frac{\partial^2 u}{\partial x^2}$$

Thus:

$$c = \sqrt{\frac{T}{\mu}}$$

3.2 From Fluid Dynamics (Acoustics)

Using:

- Continuity equation
- Euler’s equation
- Equation of state

3.2 From Fluid Dynamics (Acoustics)

Using:

- Continuity equation
- Euler's equation
- Equation of state

We obtain:

$$\frac{\partial^2 p}{\partial t^2} = c^2 \nabla^2 p$$

where p is pressure fluctuation.

---

4. General Solutions

4.1 One-Dimensional Case

4. General Solutions

4.1 One-Dimensional Case

$$\Phi(x,t) = f(x - ct) + g(x + ct)$$

This represents:

- Right-moving wave: $f(x - ct)$
- Left-moving wave: $g(x + ct)$

4.2 Higher Dimensions

4.2 Higher Dimensions

Solutions include:

- Plane waves:

$$\Phi = Ae^{i(\mathbf{k}\cdot\mathbf{r}-\omega t)}$$

- Spherical waves:

$$\Phi(r,t) = \tfrac{1}{r}f(r-ct)$$

---

5. Energy and Conservation Laws

The wave equation conserves energy:

5. Energy and Conservation Laws

The wave equation conserves energy:

$$E = \int \left[\frac{1}{2} \left(\frac{\partial \Phi}{\partial t} \right)^2 + \frac{c^2}{2} |\nabla \Phi|^2 \right] dV$$

Interpretation:

- First term → kinetic energy
- Second term → potential energy

Interpretation:

- First term → kinetic energy
- Second term → potential energy

---

6. Physical Interpretations

The nature of the wave depends on :

Field	Physical Meaning	Wave Type
Displacement	String/solid motion	Mechanical
Pressure	Sound waves	Mechanical
Velocity potential	Fluid motion	Mechanical
Scalar field	Abstract field	Non-mechanical

---

7. Relation to Relativity

7. Relation to Relativity

Rewriting:

$$\frac{1}{c^2}\frac{\partial^2 \Phi}{\partial t^2} - \nabla^2 \Phi = 0$$

This becomes:

$$\Box \Phi = 0$$

This shows:

- The classical wave equation is **Lorentz invariant**
- It naturally extends into **relativistic physics**

This shows:

- The classical wave equation is **Lorentz invariant**

- It naturally extends into **relativistic physics**

---

8. Boundary and Initial Conditions

8. Boundary and Initial Conditions

To fully determine a solution:

- Initial displacement:

$$\Phi(\mathbf{r}, 0)$$

- Initial velocity:

$$\frac{\partial \Phi}{\partial t}(\mathbf{r}, 0)$$

- Boundary conditions:
  - Fixed
  - Free
  - Periodic

---

9. Applications

9.1 Acoustics

- Sound propagation
- Noise analysis

9.2 Engineering

- Vibrations in structures
- Signal transmission

9.3 Physics

- Field theory
- Precursor to quantum wave equations

---

10. Limitations

- Assumes linearity
- Ignores dissipation
- Assumes homogeneous medium

---

11. Conclusion

The classical wave equation is a universal mathematical framework that governs wave propagation across diverse physical systems. Its simplicity conceals deep physical insight: the evolution of a field in time is determined by its spatial curvature. Whether describing sound, vibrations, or abstract fields, the wave equation remains one of the most powerful tools in theoretical and applied physics.

Future Work

- Nonlinear wave equations

- Dissipative systems
- Quantum wave generalizations
- Numerical simulations

Appendix A: Notation

Appendix A: Notation

- Φ: scalar field
- c: wave speed
- ∇^2: Laplacian operator

Appendix B: Key Insight

Wave propagation arises when spatial curvature generates temporal acceleration.

📄 Section 12: Density Field Representation in Classical Wave Propagation

12.1 Introduction

In the study of wave phenomena in continuous media, the mass density is generally a function of both space and time, expressed as:

$$\rho = \rho(x, t)$$

This dependence indicates that the medium is **compressible and dynamically evolving**, which is a necessary condition for the propagation of mechanical disturbances. Spatial and temporal variations in density are fundamental signatures of wave motion in fluids and gases.

12.2 Perturbative Decomposition of Density

To analyze wave behavior, the density field is decomposed into an equilibrium component and a perturbation:

$$\rho(x,t) = \rho_0 + \rho'(x,t)$$

where:

- ρ_0 denotes the uniform equilibrium density
- $\rho'(x,t)$ represents a small deviation from equilibrium

Under the assumption:

$$|\rho'| \ll \rho_0$$

the system can be treated within the **linear regime**, enabling analytical tractability.

12.3 Governing Wave Equation for Density Perturbations

The propagation of density disturbances in a compressible medium is governed by the fundamental conservation laws of mass and momentum, together with an appropriate equation of state. Under the assumption of small deviations from equilibrium, the governing equations may be linearized, allowing a tractable analytical description.

By eliminating the velocity field from the linearized continuity and momentum equations, one obtains a second-order partial differential equation for the density perturbation $\rho'(x,t)$. The resulting equation takes the form:

$$\frac{\partial^2 \rho'(x,t)}{\partial t^2} = c^2 \nabla^2 \rho'(x,t)$$

where c denotes the characteristic wave speed of the medium.

In three-dimensional Cartesian coordinates, this equation can be written explicitly as:

$$\frac{\partial^2 \rho'}{\partial t^2} = c^2 \left(\frac{\partial^2 \rho'}{\partial x^2} + \frac{\partial^2 \rho'}{\partial y^2} + \frac{\partial^2 \rho'}{\partial z^2} \right)$$

The wave speed c is determined by the thermodynamic properties of the medium and is given by:

$$c^2 = \left(\frac{\partial p}{\partial \rho} \right)_{\text{equilibrium}}$$

This formulation demonstrates that small density perturbations propagate as waves, with their temporal evolution governed by spatial variations. It provides a direct connection between the underlying conservation laws and the observable phenomenon of wave propagation in continuous media.

12.4 Physical Interpretation

The perturbation $\rho'(x,t)$ corresponds to localized fluctuations in density that propagate through the medium:

- Regions where $\rho' > 0$ correspond to **compressions**
- Regions where $\rho' < 0$ correspond to **rarefactions**

These alternating regions constitute a **longitudinal wave**, in which particle motion is parallel to the direction of propagation.

12.5 Coupling with Pressure and Velocity Fields

Density fluctuations are intrinsically linked to other physical variables. For small perturbations, the pressure variation satisfies:

$$p' = c^2 \rho'$$

Furthermore, the velocity field is coupled to ρ' through the continuity and momentum equations. These relationships demonstrate that wave propagation in a compressible medium is governed by a **coupled system of field variables**, rather than a single isolated quantity.

12.6 Relation to the General Scalar Wave Equation

12.6 Relation to the General Scalar Wave Equation

The classical scalar wave equation:

$$\frac{\partial^2 \Phi}{\partial t^2} = c^2 \nabla^2 \Phi$$

admits a direct physical interpretation when the scalar field is identified with the density perturbation:

$$\Phi \equiv \rho'(x, t)$$

This establishes that the wave equation provides a general mathematical framework that encompasses **physically measurable quantities**, such as density fluctuations.

12.7 Implications for Wave Theory

This formulation leads to several important conclusions:

- Wave propagation arises from **small deviations about equilibrium**
- Density perturbations serve as a **primary carrier of mechanical waves**
- The classical wave equation unifies different physical descriptions under a single mathematical structure

---

12.8 Conclusion

The representation of density as a space- and time-dependent field provides a rigorous foundation for understanding mechanical wave phenomena. By isolating the perturbative component, one obtains a quantity that satisfies the classical wave equation, thereby linking the abstract mathematical formalism to concrete physical behavior. This perspective reinforces the universality of the wave equation in describing propagating disturbances in continuous media.
